Course Information

- **Course number and title:** ME 35400: Machine Design
- **CRN:** 36659
- **Meeting day(s) and time(s)**: Async-Online.
- **Instructional Modality**: Online.
- **Course credit hours:** 3
- **Prerequisites (if any):**
  ME 26300: Introduction to Mechanical Engineering Design, Innovation, and Entrepreneurship
  ME 27400: Basic Mechanics II
  ME 32300: Mechanics of Materials

Course Description

Design, analysis, and selection of machine components for fluctuating loading. Application of design fundamentals to mechanical components and integration of components to form systems.

Learning Resources, Technology and Texts

Required Textbook:

Shigley's Mechanical Engineering Design, 2024 Release, J.K. Nisbett., R.G. Budynas, McGraw-Hill Education, New York, 2024. [ISBN: 978-1-265-47269-6 (bound)]. [ISBN: 978-1-266-73171-6 (loose-leaf)].

You are also allowed to use:
Shigley's Mechanical Engineering Design, Eleventh Edition, R.G. Budynas and J.K. Nisbett. McGraw-Hill Education, New York, 2020. [ISBN 978-0-07-339821-1].

You can use either the 2024 or 2020 version of the textbook. Either a hardcopy or electronic version of the textbook is allowed, as long as students acquire the textbook lawfully.

Please only use the US edition. Do NOT use the international edition.

Software/web resources:
Students need to have suitable software to watch the lecture videos and access other course contents in Brightspace, take exams by Honorlock, and scan/upload their completed homework and exams. "Purdue has licensed Office 365 from Microsoft to make it available for free on an individual basis to students, faculty and staff, including use on their personally owned machines." (quoted from https://it.purdue.edu/services/microsoft-office-365.php)

Use of artificial intelligence (AI) or Large Language Models (LLM):
Students are allowed to use AI tools during the learning process for work that will not be assessed (evaluated). No AI tool is allowed for assessed work (such as homework and exams).

Hardware requirements:
Students need to have electronic devices that facilitate their learning, completion and submission of homework and exams in the ways as described in this syllabus. In particular, students need to have a computer, laptop and/or other device with a camera, speaker and microphone to allow them to take exams by Honorlock.

Tutoring support:
Information on tutoring support, if available, may be posted in Brightspace.

Brightspace learning management system (LMS):
Access the course via Purdue's Brightspace learning management system. Begin with the Start Here tab, which offers further insight into the course and how you can be successful in it. It is strongly suggested that you explore and become familiar not only with the site navigation, but also with the content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Learning Outcomes

By the end of the course, you will be able to:

1. Apply fundamental concepts from statics, dynamics, and mechanics of materials to the design of machine components and/or systems.
2. Understand fatigue failure mechanisms.
3. Apply static and fatigue failure theories to the design of machine components and/or systems.
4. Select standard machine components and materials.
5. Enhance problem-solving and communication skills through course assignments.

Assignments

Homework:
Typically, one homework assignment will be given per week. The assignment will be posted in Brightspace with the due date indicated. Students will receive an email notification on the day of homework posting. Typically, students have one week to finish the homework. No homework submitted late will be accepted; but the homework assignment with the lowest grade will be dropped.

Exams:
There are two exams: Exam 1 and Exam 2. Exam 2 will mainly directly cover the topics beyond the range for Exam 1. Exams will be open-textbook.

Grading Scale

The final grades will be calculated with the following weights for each category:
Homework: 30%
Higher grade of Exams 1 and 2: 50%
Lower grade of Exams 1 and 2: 20%

The grade cutoffs for the final letter grade will not be more stringent than the following.

A+	93 – 100	A	90 – 92.99	A-	87 – 89.99
B+	83 – 86.99	B	80 – 82.99	B-	77 – 79.99
C+	73 – 76.99	C	70 – 72.99	C-	67 – 69.99
D+	63 – 66.99	D	60 – 62.99	D-	57 – 59.99
F	below 57				